MATH211
Discrete Mathematics

Set Theory

Agenda

2.1 Sets

2.2 Set Operations

Set builder notation, subset, Cartesian product, power set, union, intersection, complements, set identities

Discrete Mathematics and Its Applications

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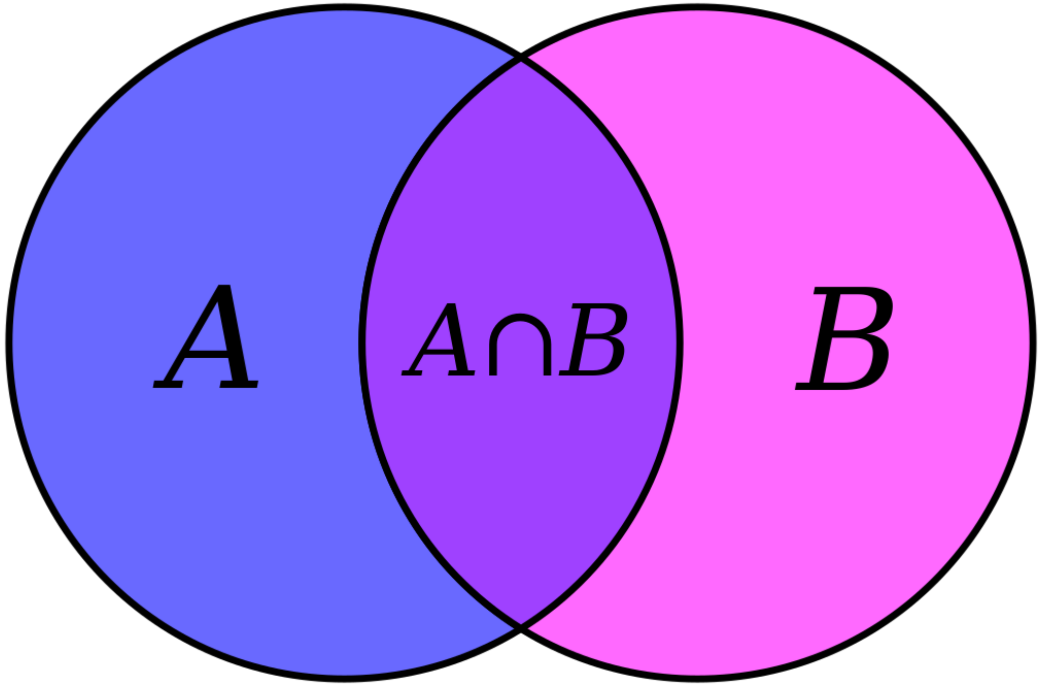


2.1. Sets

Introduction to Set Theory

- A *set* is a structure, representing an *unordered* collection (group, plurality) of zero or more *distinct* (different) objects.
  - The students in this class
  - All the video games on your phone, etc.

- The objects in a set are called the elements, members, or items of the set.

Introduction to Set Theory

$S = \{1, 2, 3\}$

- For sets, we'll use variables ***S***, ***T***, ***U***, …
- We can denote a set ***S*** in writing by *listing* all of its elements in curly braces:
  - $\{1, 2, 3\}$ is the set of whatever 3 objects are denoted by 1, 2, 3.
- We write $\mathbf{1 \in S}$ to denote that 1 is an element/member of the set S.
- The notation $\mathbf{5 \notin S}$ denotes that 5 is not an element/member of the set S.

Introduction to Set Theory

The set S of odd positive integers less than 10 can be expressed by

$$S = \{1, 3, 5, 7, 9\}.$$

The set S of positive integers less than 100 can be expressed by

$$S = \{1,2,3,\ldots,99\}.$$

Basic properties of sets

- Sets are inherently *unordered*:
  - No matter what objects a, b, and c denote,
    {a, b, c} = {a, c, b} = {b, a, c} =
    {b, c, a} = {c, a, b} = {c, b, a}

- All elements are *distinct* (unequal); multiple listings make no difference!
  - **{a, b, c} = {a, a, b, a, b, c, c, c, c}.**
  - This set contains at most 3 elements!

Basic notations for sets

- Another way to describe a set is to use **set builder** notation.
- The set S={1, 2, 3, 4} = {x | x is an integer where x>0 and x<5}
- **The set O of odd positive integers less than 10 can be expressed by**

O= {1, 3, 5, 7, 9}

$$\boxed{O = \{x \mid x \text{ is an odd positive integer less than } 10\},}$$

$$\boxed{O = \{x \in \mathbf{Z}^+ \mid x \text{ is odd and } x < 10\}.}$$

Basic notations for sets

$\mathbf{N} = \{0, 1, 2, 3, \ldots\}$, the set of all **natural numbers**
$\mathbf{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, the set of all **integers**
$\mathbf{Z}^+ = \{1, 2, 3, \ldots\}$, the set of all **positive integers**
$\mathbf{Q} = \{p/q \mid p \in \mathbf{Z}, q \in \mathbf{Z}, \text{ and } q \neq 0\}$,
the set of all **rational numbers**
$\mathbf{R}$, the set of all **real numbers**
$\mathbf{R}^+$, the set of all **positive real numbers**
$\mathbf{C}$, the set of all **complex numbers**.

Empty Set

- There is a special (unique) set that has no elements/members. This set is called **the empty set**, or **null set**, and is denoted by **ø.**

- The empty set can also be denoted by, $\varnothing = \{\ \} = \{x \mid \text{False}\}$

- No matter the domain of discourse, we have the axiom $\neg\exists x$ such that $x \in \varnothing$

Definition of Set Equality

- Two sets are declared to be equal *if and only if* they contain exactly the same elements.
- In particular, it does not matter *how the set is defined or denoted.*
- Two sets, A and B, are equal:

$$A = B \Leftrightarrow \forall x(x \in A \leftrightarrow x \in B)$$

Set equality written mathematically

- If A and B are sets, then A and B are equal if and only if **$\forall x(x \in A \leftrightarrow x \in B)$.** We write $A=B$, if A and B are equal sets.

- The sets **{1,3,5} and {3,5,1}** are equal, because they have the same elements.

- **{1,3,3,5,5,5}** is the same as the set **{1,3,5}** because they have the same elements.

Cardinality (i.e., size) of a Finite set

- The cardinality of a set S is the number of distinct/different elements in S.
- The cardinality of S is denoted by **|S|**
- If there are **n** distinct elements in the set **S** where **n** is a nonnegative integer, we say that **S** is a **finite set.**
- **|S| = n, where n is the number of distinct elements in the set**
- Example: **S = {a, a, b, a, b, c, c, c, c}, |S| = 3.**

Cardinality (i.e., size) of an Infinite set

- A set is said to be infinite if it is not finite. (i.e., we can't count the number of its elements)
- Conceptually, sets may be *infinite* (*i.e.*, not *finite*, without end, unending).
- Symbols for some special infinite sets:
  N = {0, 1, 2, …} The **N**atural numbers.
  Z = {…, -2, -1, 0, 1, 2, …} The Integers.
  Z+ = {1, 2, 3, 4, …} The positive **I**ntegers.
  R = The "**R**eal" numbers, such as 374.1828…
- **The cardinality of an infinite set is infinity**, but some infinities are bigger than others.

Example

$$S = \{a, b, c, d\}$$

$$A = \{1, 2, 3, 7, 9\}$$

$$\emptyset = \{\ \}$$

$$S = \{a, b, c, d\}$$
$$|S| = 4$$

$$A = \{1, 2, 3, 7, 9\}$$
$$|A| = 5$$

$$\emptyset = \{\ \}$$
$$|\emptyset| = 0$$

Basic Set Relations: Member of

- $x \in S$ ("x is in S") is the proposition that object x is an ***element*** or ***member of the*** set S *[belong Rule]*.
  - **e.g. 3∈N**
- $x \notin S :\equiv \neg(x \in S)$ "x is not in S" [not belong Rule]
  - **e.g. *3.5 ∉ Z***

Subset and Superset Relations

- The set **S** is said to be a subset of **T** if and only if every element of S is also an element of T.

- We use the notation **S⊆T** to indicate that S is a subset of the set T.

- $\mathbf{S \subseteq T} = \boldsymbol{T \supseteq S} \Leftrightarrow \forall \mathbf{x}\ (\mathbf{x} \in \mathbf{S} \rightarrow \mathbf{x} \in \mathbf{T})$**.** All elements of S are found in T. But T can have extra elements and might be equal to S too.

Subset and Superset Relations

- $\emptyset \subseteq S$ (The empty set is a subset of any set)
- $S \subseteq S$ (Any set is technically a subset of itself)
- $S \supseteq T$ ("S is a superset of T") means $T \subseteq S$.
- Note $S=T \Leftrightarrow S \subseteq T \wedge T \subseteq S \Leftrightarrow S \subseteq T \wedge S \supseteq T$
- $S \nsubseteq T$ means $\neg(S \subseteq T)$, i.e. $\exists x(x \in S \wedge x \notin T)$

Subset and Superset Relations

$S \subseteq T \Leftrightarrow \forall x\, (x \in S \rightarrow x \in T)$

T= {1, 2, 4, 5, 7, 8, 10}
S= {2, 4, 7}

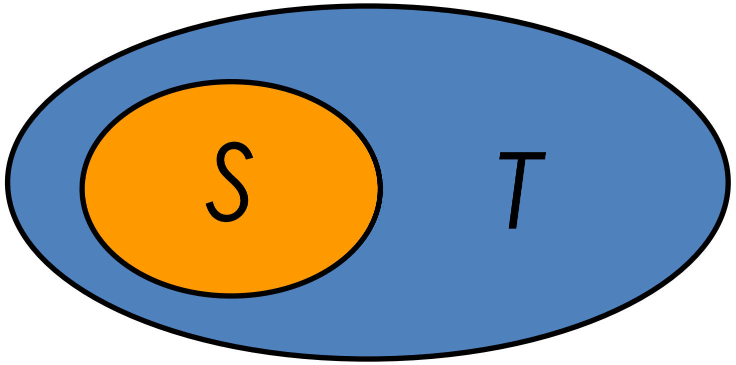
$S \subseteq T$
$T \subseteq S$ **Wrong !!**
$T \supseteq S$

T= {1, 2, 4, 5, 7, 8, 10}
S= {1, 2, 4, 5, 7, 8, 10}

$S \subseteq T$
$T \subseteq S$
$S = T \Leftrightarrow S \subseteq T \wedge T \subseteq S$

Proper (Strict) Subsets & Supersets

- If set S is a subset of the set T but that $S \neq T$, we write $S \subset T$ and say that S is a **proper** (strict) **subset** of T.
- $\mathbf{S \subset T \Leftrightarrow \forall x(x \in S \rightarrow x \in T) \wedge \exists x(x \in T \wedge x \notin S)}$
- $S \subset T$ ("S is a **proper subset** of T") means that $S \subseteq T$ but $T \nsubseteq S$. Similar for $S \supset T$.
- Example:
  - $\{1,2\} \subset \{1,2,3\}$



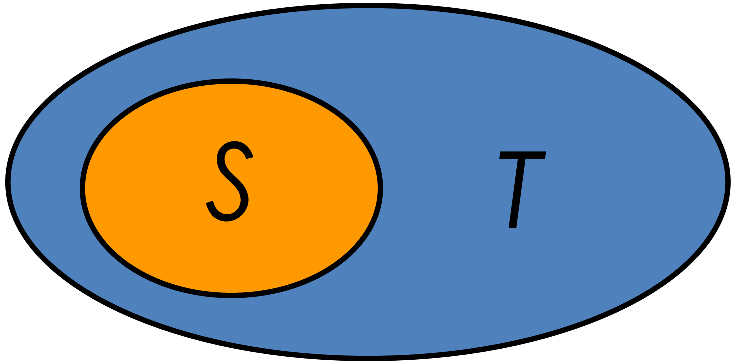
Venn Diagram equivalent of $S \subset T$

Proper (Strict) Subsets & Supersets

$S \subset T \Leftrightarrow \forall x(x \in S \rightarrow x \in T) \wedge \exists x(x \in T \wedge x \notin S)$, All elements of S are in T But $S \neq T$ so T must have at least one extra element.

$T = \{1, 2, 4, 5, 7, 8, 10\}$
$S = \{2, 4, 7\}$

$S \subset T$
Equivalent to
$S \subseteq T$ and $T \nsubseteq S$



Venn Diagram equivalent of $S \subset T$

Proper (Strict) Subsets & Supersets

Which one of the following statements is wrong?

1. $\{1,2\} \subseteq \{1,2,3\}$
2. $\{1,2\} \subseteq \{1,2\}$
3. $\{1,2\} \subset \{1,2,3\}$
4. $\{1,2\} \subset \{1,2\}$

Proper (Strict) Subsets & Supersets

1. $1, 2 \subseteq 1, 2, 3$ ✅
   - True: {1,2} is a subset of {1,2,3}
2. $1, 2 \subseteq 1, 2$ ✅
   - True: Every set is a subset of itself
3. $1, 2 \subset 1, 2, 3$ ✅
   - True: {1,2} is a **proper subset** of {1,2,3} (not equal)
4. $1, 2 \subset 1, 2$ ❌
   - **False:** A set **is not** a proper subset of itself — this is the incorrect one.

The *Power Set* Operation

- The *power set* of a set S is the set of all subsets of S *and it is denoted by* $P(S)$

$$P(S) = \{x \mid x \subseteq S\}$$

- Sometimes $P(S)$ is written 2^S
- The number of elements in the power set of a finite set is denoted by $\mathbf{|P(S)| = 2^{|S|}}$
- It turns out $\forall S\colon |P(S)| > |S|$, e.g. $|P(\mathbb{N})| > |\mathbb{N}|$
- *There are different sizes of infinite sets!*

The *Power Set* Operation

$$S = \{1,2,3\}$$

$$|P(S)| = 2^3 = 8 \text{ elements}$$

$$P(S) = 2^S$$

$$= \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$$

Sets Are Objects, Too!

- Can make a SET OF SETS!!!
- The objects that are elements of a set may *themselves* be sets.
- *E.g.* let $S=\{x \mid x \subseteq \{1,2,3\}\}$
  then $S=\{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$

IMPORTANT Note that $1 \neq \{1\} \neq \{\{1\}\}$!!!!

1	number 1
$\{1\}$	set containing element "1"
$\{\{1\}\}$	SET OF SETS

Sets Are Objects, Too!

$S = \{a, b, c, d, \{2\}\}$
$|S| =$

$A = \{1, 2, 3, \{2,3\}, 9\}$
$|A| =$

$S = \{a, b, c, d, \{2\}\}$
$|S| = 5$

$A = \{1, 2, 3, \{2,3\}, 9\}$
$|A| = 5$

Ordered *n*-tuples

- N-tuples are like sets, except that duplicates matter, and the order makes a difference.
- For $n \in \mathbf{N}$, an *ordered n-tuple* is written $(a_1, a_2, \ldots, a_n)$. The ordered n tuple $(a_1, a_2, \ldots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element, … , and a_n as its nth element.
- In particular, ordered 2-tuples are called ordered pairs (e.g., the ordered pairs (a,b))
- **Note** $(1, 2) \neq (2, 1) \neq (2, 1, 1)$.

Cartesian Products of Sets

- Let A and B be sets. The Cartesian product of A and B, denoted by $A \times B$, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$.
- For sets A, B, their Cartesian product, $A \times B :\equiv \{(a, b) \mid a \in A \wedge b \in B\}$
- For sets B, A, their Cartesian product, $B \times A :\equiv \{(b, a) \mid b \in B \wedge a \in A\}$
- E.g., A = {a, b}, and B = {1, 2}

  A x B = {(a,1),(a,2),(b,1),(b,2)} (a set of ordered pairs)

  B x A = {(1,a),(2,a),(1,b),(2,b)} (a set of ordered pairs)

Cartesian Products of Sets

Remarks

- For finite sets A, B, $|A \times B| = |A|\,|B|$.
- The Cartesian product is ***not*** commutative:

$$\forall AB: A \times B \neq B \times A$$

- Extends to multiple sets

$$A_1 \times A_2 \times \ldots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i \text{ for } i \in \{1, 2, \ldots n\}\}$$

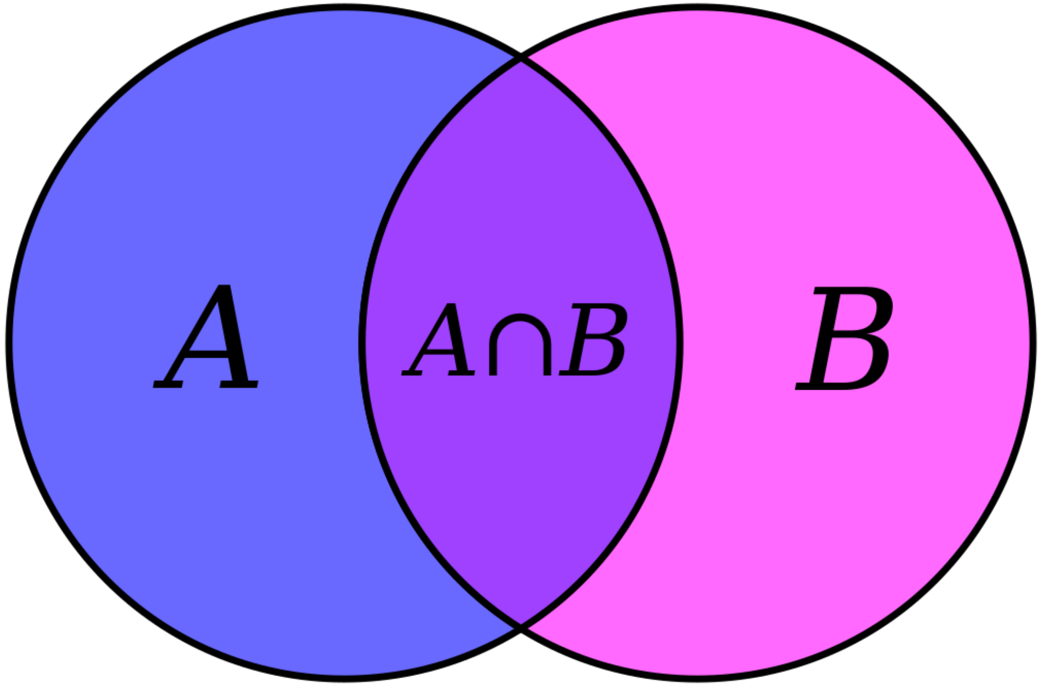
Cartesian Products of Sets

Let $A = \{1,2\}$, and $B = \{a, b, c\}$

$|A \times B| = |A| * |B| = 2 * 3 = 6$

$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$

$\boxed{\text{Find } B \times A\,?}$

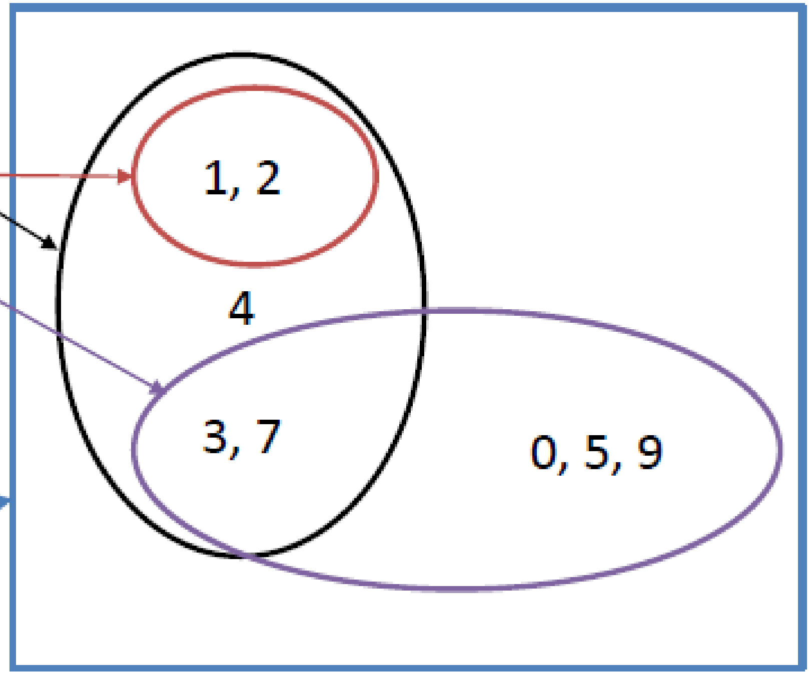


2.2 Set Operations

Venn Diagrams

- Geometric representation of sets

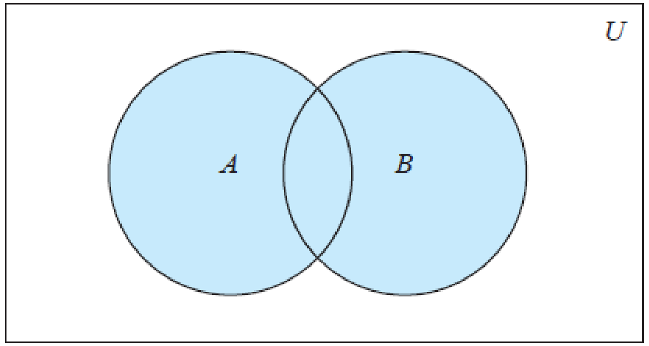
$$A = \{1,2,3,4,7\}$$
$$B = \{0,3,5,7,9\}$$
$$C = \{1,2\}$$



Universal Set

The Union Operator

- For sets *A*, *B*, their *union* $(A\cup B)$ is the set containing all elements that are either in *A*, **or** ("$\vee$") in *B* (or, of course, in both)



- Formally, $\forall A,B: A\cup B = \{x \mid x\in A \vee x\in B\}$

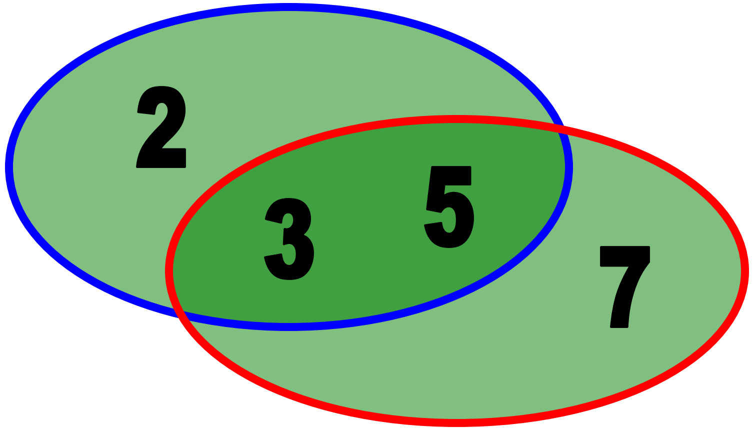
- Note that $A\cup B$ contains all the elements of A and also all the elements of B

- $A\cup B$ is a **superset** of both *A* and *B* (in fact, it is the smallest such superset):

$$\forall A,B: (A \subseteq A\cup B) \wedge (B \subseteq A\cup B)$$

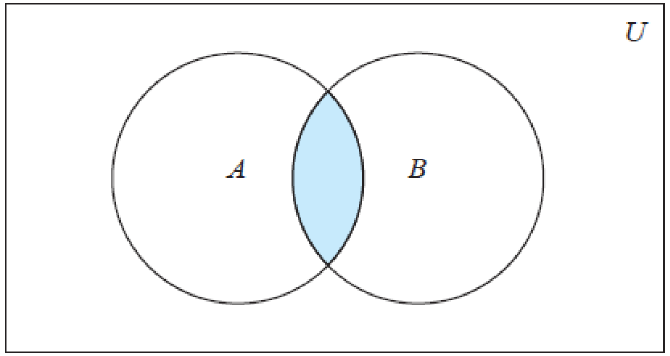
Union Examples

- $\{a,b,c\} \cup \{2,3\} = \{a,b,c,2,3\}$
- $\{2,3,5\} \cup \{3,5,7\} = \{2,3,5,3,5,7\} = \{2,3,5,7\}$



The Intersection Operator

- For sets *A*, *B*, their *intersection* $(A \cap B)$ is the set containing all elements that are in both *A* **and** ("$\wedge$") in *B*.
- Formally, $\forall A, B: A \cap B \equiv \{x \mid x \in A \wedge x \in B\}$

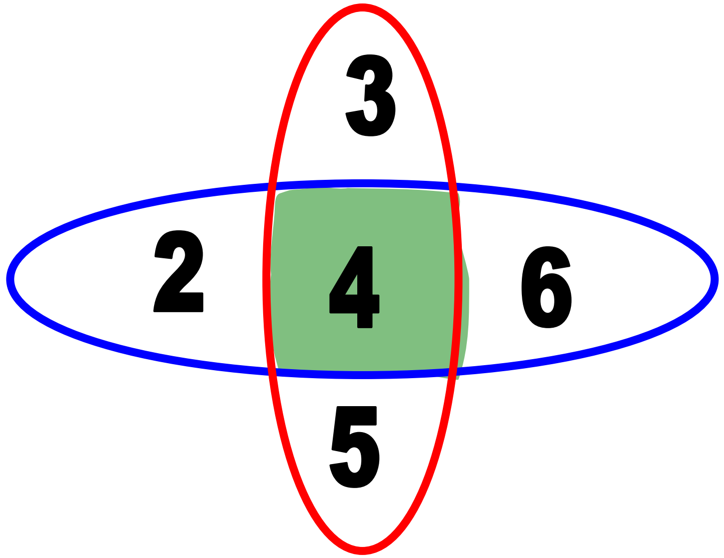


- **Remark:** $A \cap B$ is a subset of *A* **and** it is a subset of *B*:

$$\forall A, B: (A \cap B \subseteq A) \wedge (A \cap B \subseteq B)$$

Intersection Examples

- $\{a,b,c\} \cap \{2,3\} = \varnothing$
- $\{2,4,6\} \cap \{3,4,5\} = \{4\}$



Disjointedness

- Two sets A, B are called *disjoint (i.e., unjoined)* if and only if their intersection is empty.
- $(\boldsymbol{A \cap B = \varnothing})$
- Example: the set of even integers is disjoint with the set of odd integers.

Set Difference

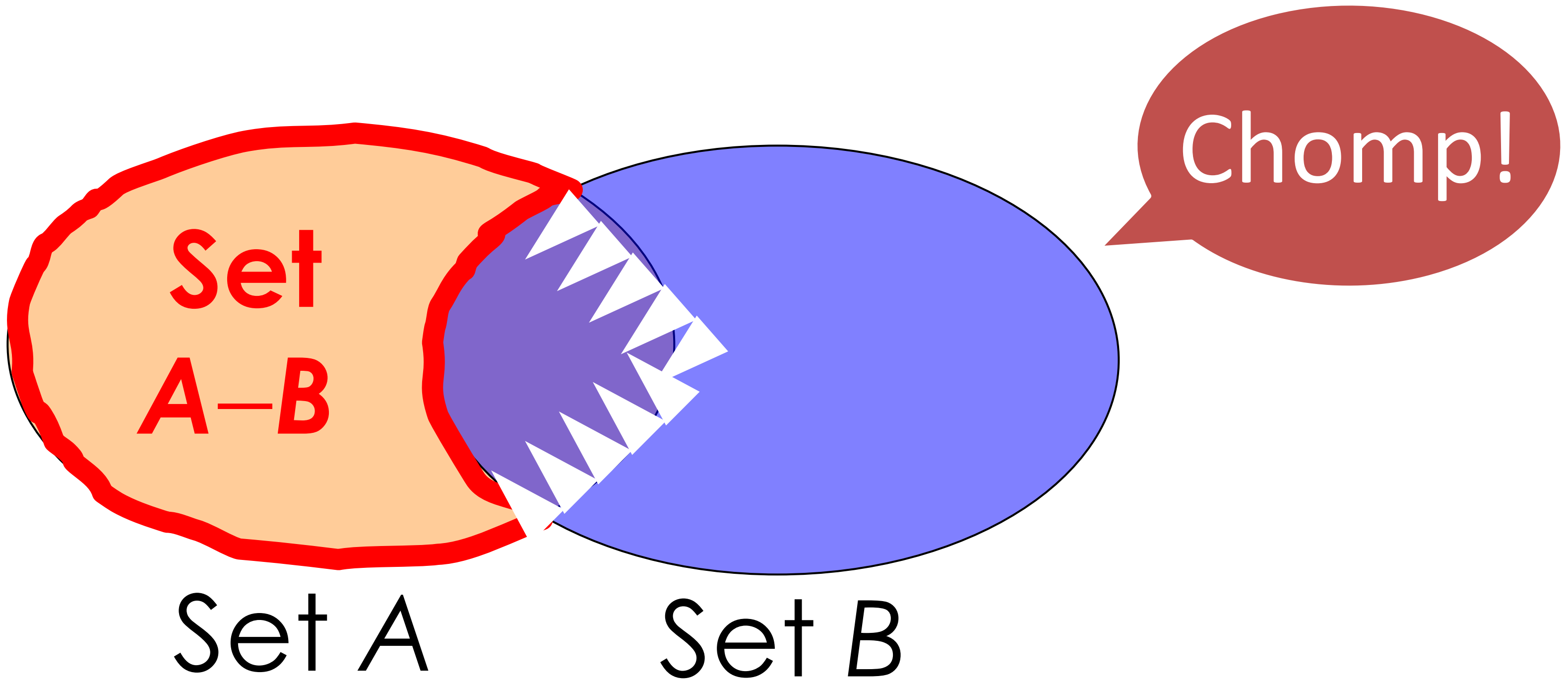
- For sets *A*, *B*, the ***difference*** *of A and B*, written ***A*-*B***, is the set of all elements that are in *A* but not in *B*.

- $A - B :\equiv \{x \mid x \in A \land x \notin B\}$

- Also called:
  The *complement of B with respect to A*.

Set Difference - Venn Diagram

- *A*-*B* is what's left after *B* "takes a bite out of *A*"



Set Difference Examples

- $\{1,2,3,4,5,6\} - \{2,3,5,7,9,11\} = \{1,4,6\}$
- $\mathbf{Z} - \mathbf{N} = \{\ldots, -1, 0, 1, 2, \ldots\} - \{0, 1, \ldots\}$

  $= \{x \mid x \text{ is an integer but not a nat. \#}\}$

  $= \{x \mid x \text{ is a negative integer}\}$

  $= \{\ldots, -3, -2, -1\}$

Set Complements

- The *universe of discourse (universal set)* can itself be considered a set, call it *U*.

- The *complement* of *A*, written $\overline{A}$, is the complement of *A* w.r.t. *U*, *i.e.*, it is *difference of U and A*, ***U - A.***

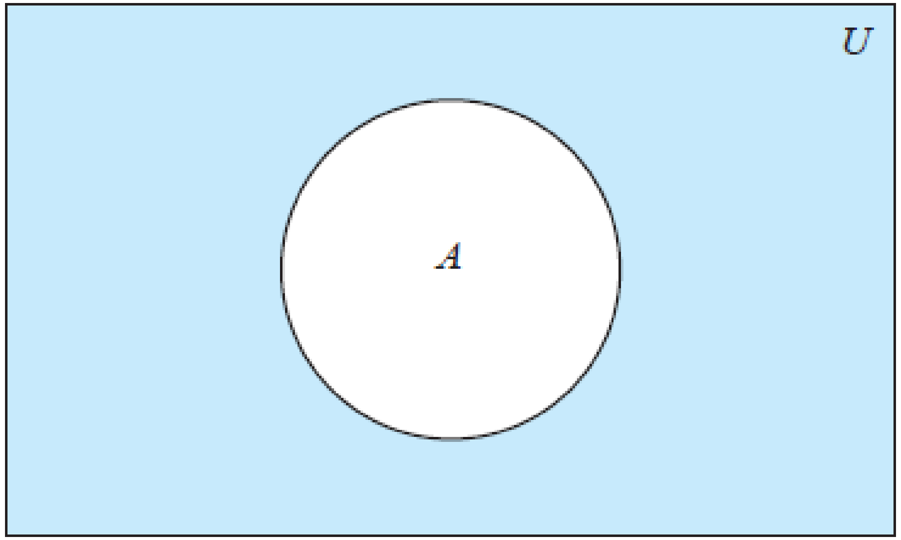
- If U={1, 2, 3, 4, 5} and A={1, 3}, then $\overline{A}$= {2, 4, 5}

- If *U*=**N**, *and A*= {3, 5}, *then* $\overline{A}$= {0, 1, 2, 4, 6, 7, …}

More on Set Complements

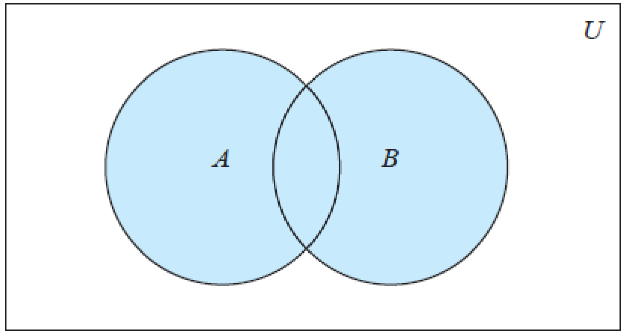
- An equivalent definition, when *U* is clear:

$$\overline{A} = \{x \mid x \notin A\}$$

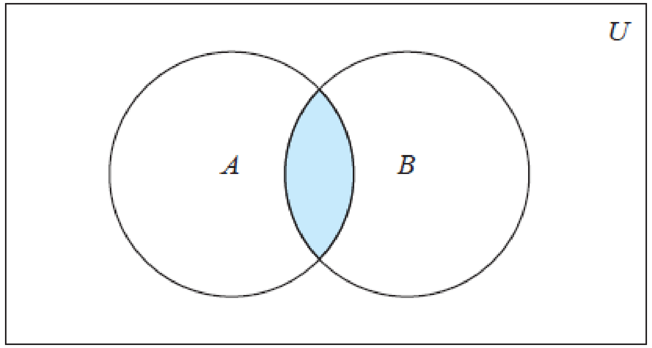


$\overline{A}$ is shaded.

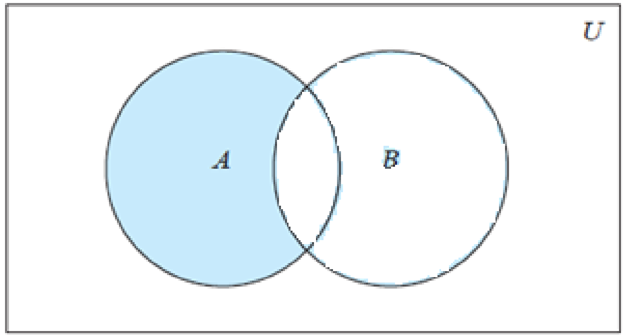
Venn Diagrams



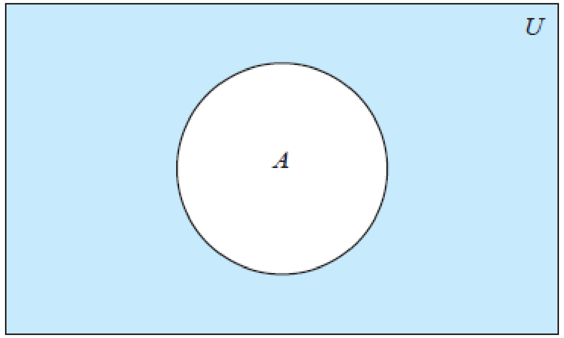
$A \cup B = \{x \mid x \in A \vee x \in B\}$



$A \cap B \equiv \{x \mid x \in A \wedge x \in B\}$



$A - B :\equiv \{x \mid x \in A \wedge x \notin B\}$



$\overline{A} = \{x \mid x \notin A\}$

DeMorgan's Law for Sets

- Exactly analogous to (and derivable from) DeMorgan's Law for propositions.

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

Set Identities

Set identities are rules that describe how sets behave under operations like union (∪), intersection (∩), and complement. They're like algebra rules, but for sets.

Identity	*Name*
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws

Set Identities

$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ $\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement laws

Proving Set Identities

There are four techniques/methods to prove statements about sets, of the form $E_1 = E_2$ (where Es are set expressions):

1. Use Mutual subsets. Prove $E_1 \subseteq E_2$ and $E_2 \subseteq E_1$ separately.
2. Use the logical equivalences [Set Identities].
3. Use the membership table.
4. Use the Ven diagram

Method 1: Mutual subsets

- Recall that to prove such identity, we must show that:
  1. The left-hand side is a subset of the right-hand side
  2. The right-hand side is a subset of the left-hand side
  3. Then conclude that the two sides are thus equal

Method 1- Mutual subsets: Example 1

Prove that $A \cup (A \cap B) = A$

Method 1- Mutual subsets: Example 1

- **Solution:** To prove $A \cup (A \cap B) = A$, we need to show that each side of the equation is a subset of the other.
- **Show that $A \cup (A \cap B)$ is a subset of A:**
  - Assume $x \in A \cup (A \cap B)$ and show that $x \in A$
  - Since $x \in A \cup (A \cap B)$.
  - Then $x \in A$ or x is in $(A \cap B)$.
  - If $x \in A$, we're done.
  - If $x \in (A \cap B)$, then $x \in A$ and $x \in B$, so $x \in A$.
  - Thus, in both cases, $x \in A$.

Method 1- Mutual subsets: Example 1

- **Show that A is a subset of A $\cup$ (A $\cap$ B):**
  - Assume $x \in A$ and show that $x \in A \cup (A \cap B)$
  - Since $x \in A$.
  - Then, It's also true that $x \in A \cup (A \cap B)$ because we're taking the union with A.
  - Therefore, $x \in$ A union $(A \cap B)$.

- Since both directions are proven, **A $\cup$ (A $\cap$ B) is a subset of A and A is a subset of A $\cup$ (A $\cap$ B)**, we have $A \cup (A \cap B) = A$.

Method 1- Mutual subsets: Example 2

Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Method 1- Mutual subsets: Example 2

- **Show that $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$,**
  - Assume $x \in A \cap (B \cup C)$ and show that $x \in (A \cap B) \cup (A \cap C)$.
  - Since $x \in A \cap (B \cup C)$, then $x \in A$, and either $x \in B$ or $x \in C$.
    - Case 1: $x \in B$. Then $x \in A \cap B$, so $x \in (A \cap B) \cup (A \cap C)$.
    - Case 2: $x \in C$. Then $x \in A \cap C$, so $x \in (A \cap B) \cup (A \cap C)$.
  - Therefore, $x \in (A \cap B) \cup (A \cap C)$.
  - Therefore, $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

Method 1- Mutual subsets: Example 2

- **Show that $(A\cap B)\cup(A\cap C) \subseteq A\cap(B\cup C)$**
- Assume $x\in (A\cap B)\cup(A\cap C)$ and show that $x\in A\cap(B\cup C)$
- Since $x\in (A\cap B)\cup(A\cap C)$, then $x\in(A\cap B)$ or $x\in(A\cap C)$
    - Case 1: $(A\cap B)$. Then $x\in A$ and $x\in B$, so $x\in A\cap(B\cup C)$
    - Case 2: $(A\cap C)$. Then $x\in A$ and $x\in C$, so $x\in A\cap(B\cup C)$
  - Therefore, $x\in A\cap(B\cup C)$
  - Therefore, $(A\cap B)\cup(A\cap C) \subseteq A\cap(B\cup C)$
- **Since $A\cap(B\cup C)\subseteq(A\cap B)\cup(A\cap C)$ and $(A\cap B)\cup(A\cap C) \subseteq A\cap(B\cup C)$. Then $A\cap(B\cup C)=(A\cap B)\cup(A\cap C)$**

Method 2: Logical Equivalence Example 1

Let A, B, and C be sets. Show that

$$\overline{A \cup (B \cap C)} = (\overline{C} \cup \overline{B}) \cap \overline{A}.$$

$$\begin{aligned}
\overline{A \cup (B \cap C)} &= \overline{A} \cap (\overline{B \cap C}) && \text{by the first De Morgan law} \\
&= \overline{A} \cap (\overline{B} \cup \overline{C}) && \text{by the second De Morgan law} \\
&= (\overline{B} \cup \overline{C}) \cap \overline{A} && \text{by the commutative law for intersections} \\
&= (\overline{C} \cup \overline{B}) \cap \overline{A} && \text{by the commutative law for unions.}
\end{aligned}$$

Method 3: Membership Tables

- Membership tables (also called truth tables for sets) are a way to show how set operations work, using 0s and 1s to indicate membership.
- Symbols:
  - **1** = element **is in** the set
  - **0** = element **is not in** the set
- Columns for different set expressions
- Rows for all combinations of memberships in constituent sets
- Prove equivalence with identical columns.

Method 3 - Membership Table

A	B	$A \cap B$	$A \cup B$	$\bar{A}$	$\bar{B}$	A-B	B-A
0	0	0	0	1	1	0	0
0	1	0	1	1	0	0	1
1	0	0	1	0	1	1	0
1	1	1	1	0	0	0	0

Method 3 - Membership Table Example 1

Prove that $(A \cup B) - B = A - B$.

A	B			
0	0			
0	1			
1	0			
1	1			

Method 3 - Membership Table Example 1

Prove that $(A \cup B) - B = A - B$.

A	B	$A \cup B$	$(A \cup B) - B$	$A - B$
0	0	0	0	0
0	1	1	0	0
1	0	1	1	1
1	1	1	0	0

Method 3 - Membership Table Example 2

Prove that $\overline{A \cap B \cap C} = \bar{A} \cup \bar{B} \cup \bar{C}$

A	B	C	$A \cap B \cap C$	$\overline{A \cap B \cap C}$	$\bar{A}$	$\bar{B}$	$\bar{C}$	$\bar{A} \cup \bar{B} \cup \bar{C}$
0	0	0						
0	0	1						
0	1	0						
0	1	1						
1	0	0						
1	0	1						
1	1	0						
1	1	1						

Method 3 - Membership Table Example 2

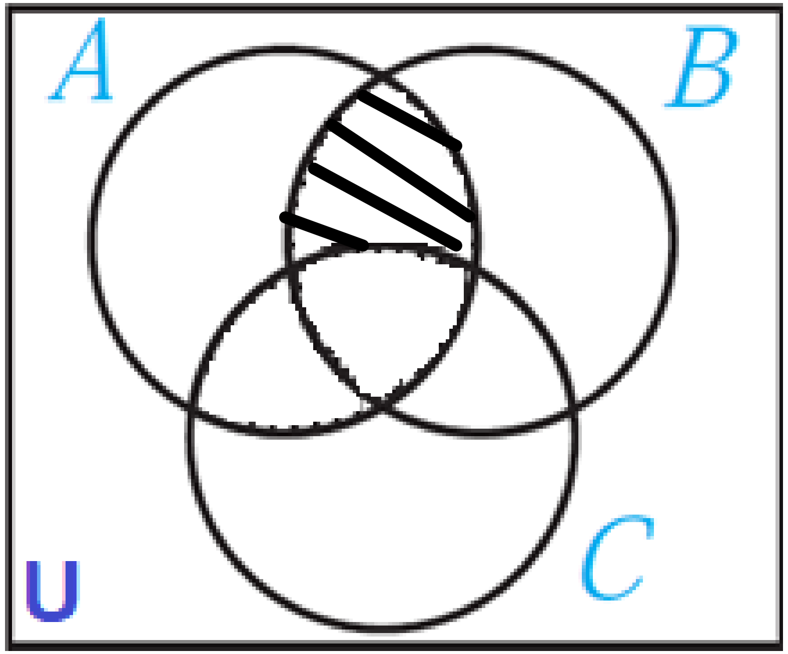
A	B	C	$A \cap B \cap C$	$\overline{A \cap B \cap C}$	$\bar{A}$	$\bar{B}$	$\bar{C}$	$\bar{A} \cup \bar{B} \cup \bar{C}$
0	0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	0	1
0	1	0	0	1	1	0	1	1
0	1	1	0	1	1	0	0	1
1	0	0	0	1	0	1	1	1
1	0	1	0	1	0	1	0	1
1	1	0	0	1	0	0	1	1
1	1	1	1	0	0	0	0	0

- 1 under a set indicates that "an element is in the set"
- If the columns are equivalent, we can conclude that indeed the two sets are equal

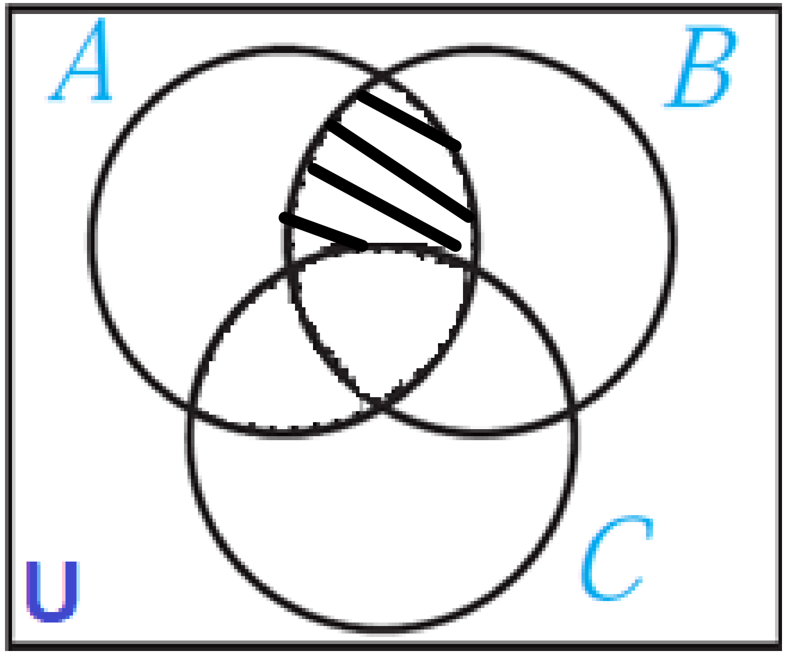
Method 4 - Venn Diagrams

Prove that A ∩ (B − C) = (A ∩ B) − C

A ∩ (B − C)



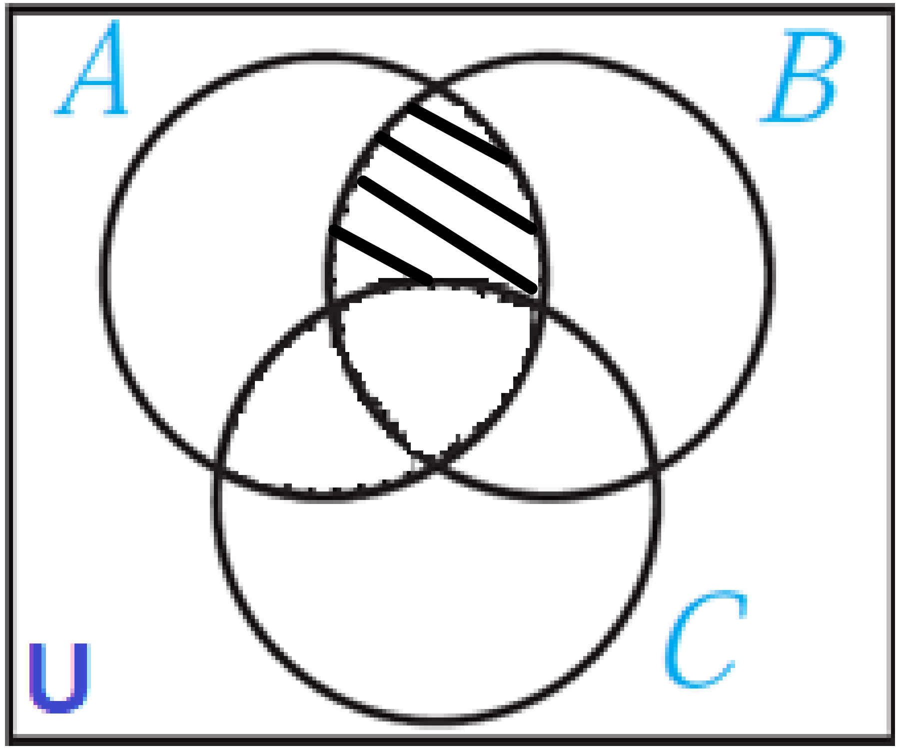
(A ∩ B) − C



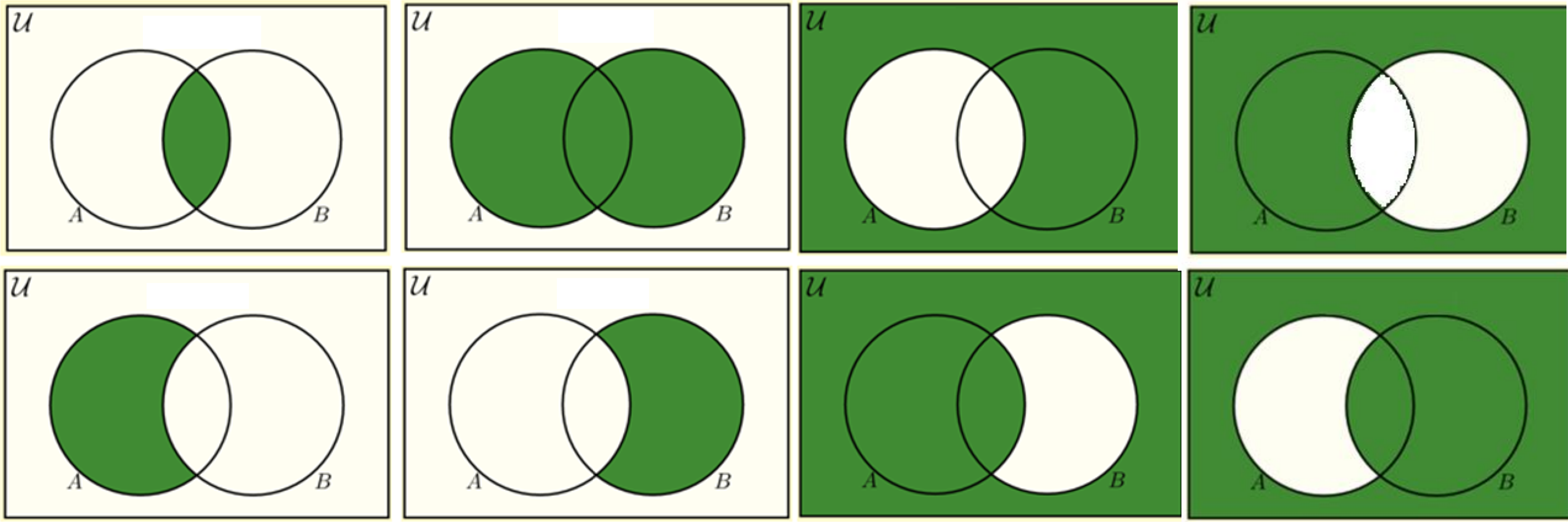
Venn Diagrams

Draw the Venn diagram for the following combination of the sets A, B, and C.

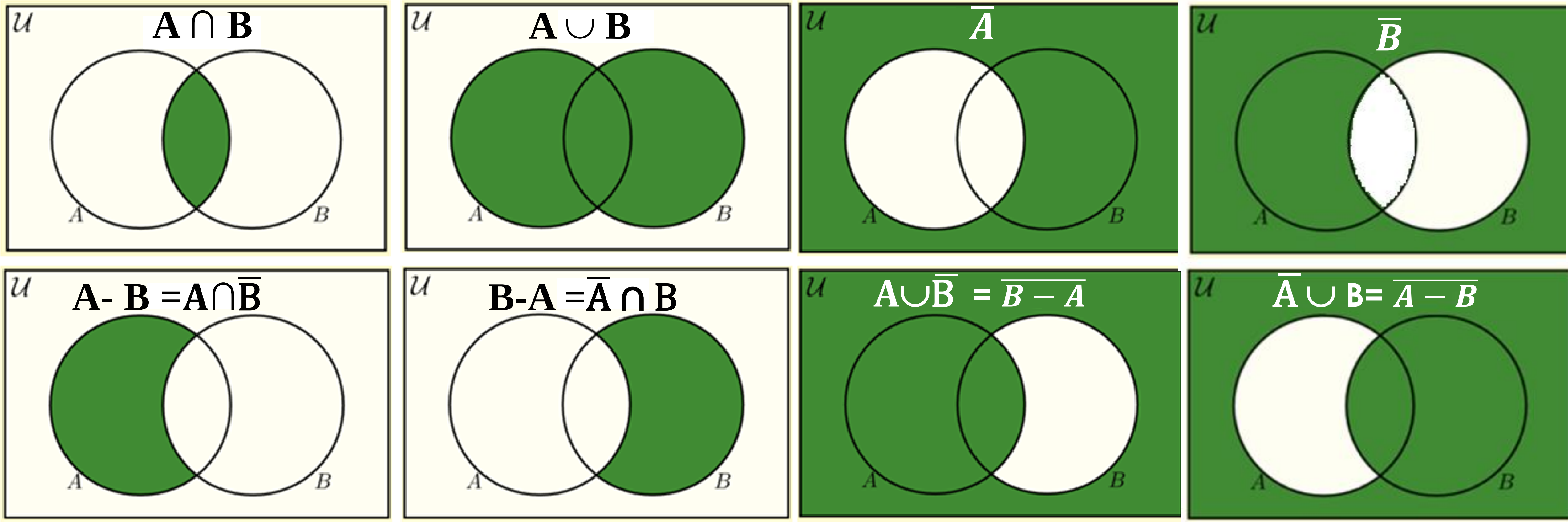
- $\mathbf{A \cap (B - C)}$



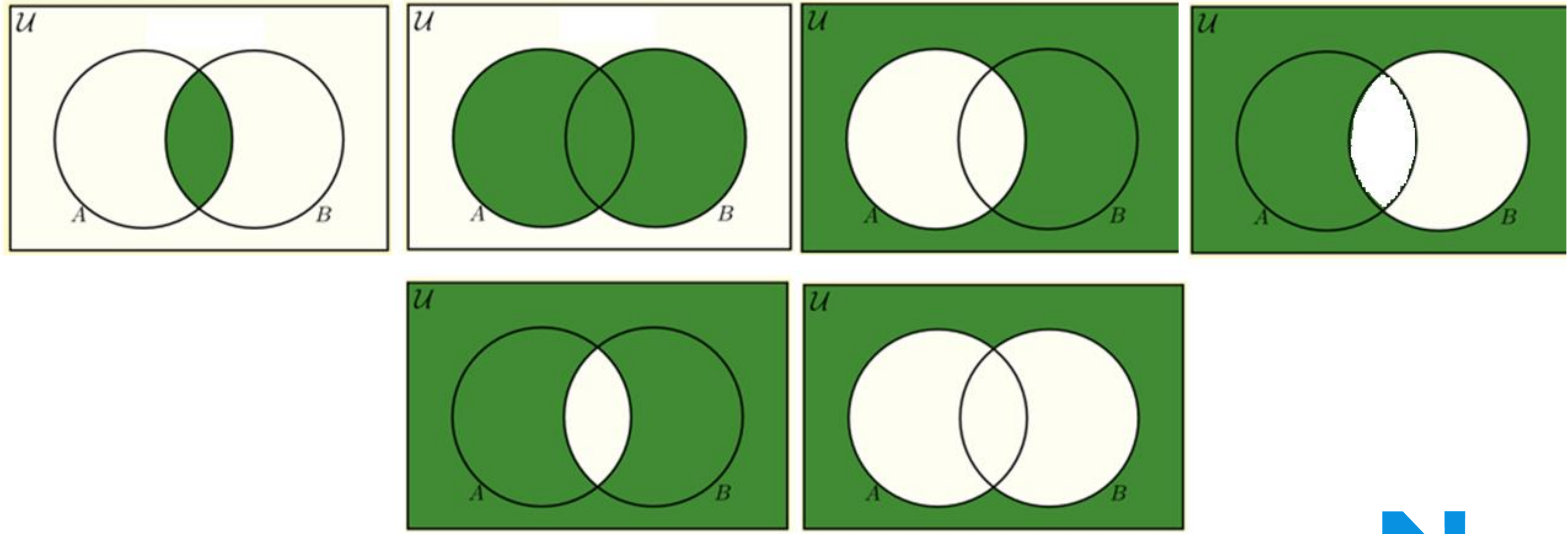
Venn Diagrams
Find the relation of the sets A and B for each of the following diagrams



Venn Diagrams
Find the relation of the sets A and B for each of the following diagrams

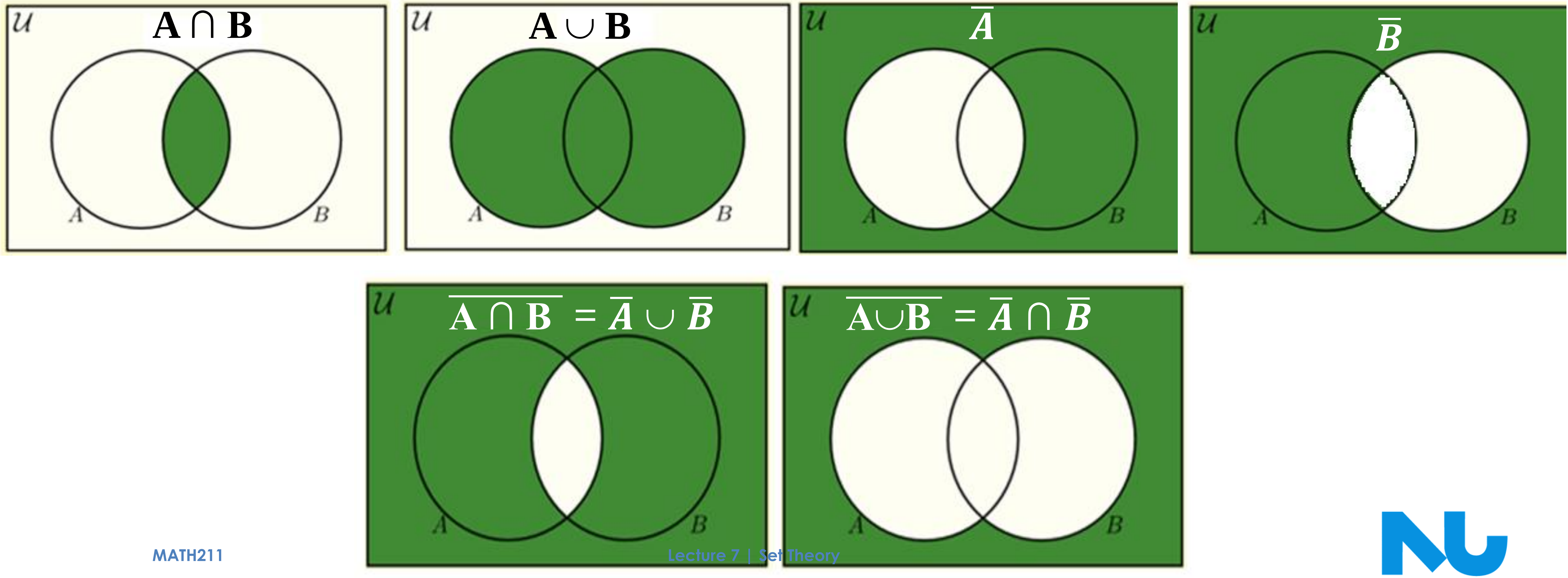


Venn Diagrams
Find the relation of the sets A and B for each of the following diagrams



Venn Diagrams

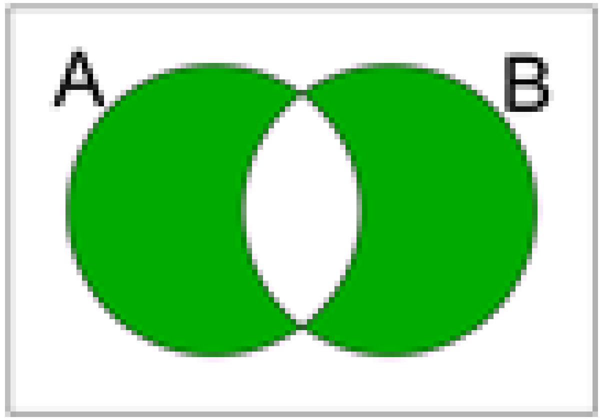
Find the relation of the sets A and B for each of the following diagrams



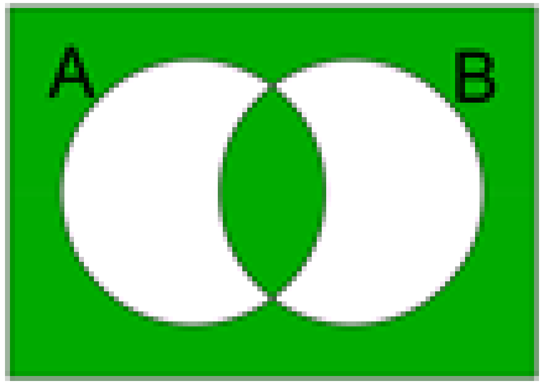
Review Questions

Problem 1

Find the relation of the sets A and B for each of the following diagrams



$(A \cap B') \cup (A' \cap B)$



$(A' \cap B') \cup (A \cap B)$

Problem 2

Prove (A∪B)–C = (A–C)∪(B–C) using membership tables.

A	B	C	$A \cup B$	$(A \cup B) - C$	$A - C$	$B - C$	$(A - C) \cup (B - C)$
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

Supplementary Material

- Discrete Math - 2.1.1 Introduction to Sets
- Discrete Math - 2.1.2 Set Relationships
- Discrete Math - 2.2.1 Operations on Sets
- Discrete Math - 2.2.2 Set Identities
- Discrete Math - 2.2.3 Proving Set Identities

Next Lecture

- Functions

See you next lecture